



Name: _____ Student No.: _____ Course/Section: _____ Date: _____



LEARNING OBJECTIVES

After completing this topic, students should be able to:

- 1 Define sampling distribution.
- 2 Explain how repeated samples form a distribution.
- 3 Describe the sampling distribution of the mean.
- 4 Explain the relationship between sampling distribution and the Law of Large Numbers (LLN).
- 5 Describe the general shape of the sampling distribution.



A. CONCEPT CHECK

Write the letter of the term that best matches each description.

- _____ 1. The distribution of a statistic (such as the sample mean) from many repeated samples of the same size taken from the same population.
- _____ 2. Taking many samples of the same size from the same population.
- _____ 3. The average of all values in the population.
- _____ 4. A subset of the population selected for a study.
- _____ 5. The average of values in a sample.

Choices

- | |
|--------------------------|
| A. Population mean |
| B. Sample |
| C. Sample mean |
| D. Repeated sampling |
| E. Sampling distribution |



B. ARRANGE THE PROCESS

Arrange the steps in the correct order to show how a sampling distribution is formed.

Forms a distribution of sample means.

Each sample produces a sample mean.

Take many random samples of the same size from the same population.

Start with one population of interest.

Order: 1) _____ 2) _____ 3) _____ 4) _____



C. FILL IN THE BLANKS

Fill in the blanks with the correct words.

1. A sampling distribution is the distribution of a _____ (such as the sample mean) obtained from many _____ samples of the same size taken from the _____ population.
2. The sample means from different samples are usually _____.
3. When the number of samples increases, the distribution of the sample means becomes more _____.



D. GUIDED EXAMPLE (Answer Provided)

A population has the numbers 2, 4, 6, 8, 10. Take all possible samples of size $n = 2$ without replacement and find the sample mean for each.

Samples and means:

(2,4) → 3 (2,6) → 4 (2,8) → 5 (2,10) → 6
(4,6) → 5 (4,8) → 6 (4,10) → 7 (6,8) → 7
(6,10) → 8 (8,10) → 9

The sampling distribution of the mean:

3, 4, 5, 5, 6, 6, 7, 7, 8, 9





A. REPEATED SAMPLING IN ACTION

A population has the scores below.

A student takes many random samples of size $n = 3$ (without replacement) and computes the sample mean for each sample.

Population scores:

2 4 6 8 10

Sample	1	2	3	4	5	6	7	8	9	10
Sample ($n = 3$)	{2, 4, 6}	{2, 4, 8}	{2, 4, 10}	{2, 6, 8}	{2, 6, 10}	{2, 8, 10}	{4, 6, 8}	{4, 6, 10}	{4, 8, 10}	{6, 8, 10}
Sample mean ($\bar{x}$)										

- Find the sample mean for each sample above and fill in the table.
- List all the sample means in order from least to greatest. _____
- What is the population mean (μ)? _____
- What do you notice about the sample means compared to the population mean?



Think!

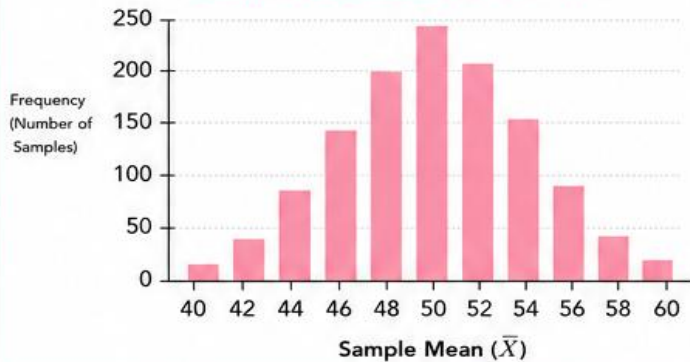
Even though the samples are different, the sample means tend to cluster around the population mean. Why?



B. SAMPLING DISTRIBUTION OF THE MEAN

The histogram below shows the sampling distribution of the sample mean $\bar{X}$ based on 1,000 repeated random samples of size $n = 5$ from a population with mean $\mu = 50$ and standard deviation $\sigma = 12$.

Sampling Distribution of $\bar{X}$ ($n = 5$)



- What is the center (mean) of the sampling distribution?

- What is the approximate shape of the sampling distribution?

- Estimate the spread (standard deviation) of the sampling distribution using the histogram. (Hint: look at where most of the data lie.)

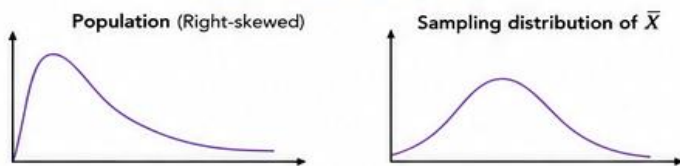
- Why does the sampling distribution of the mean have less variability than the population?



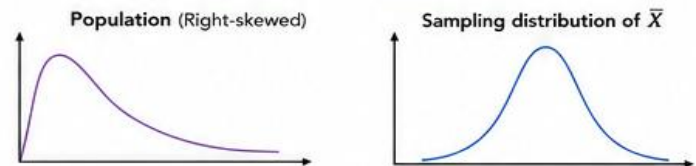
C. SHAPE OF THE SAMPLING DISTRIBUTION

A population is right-skewed. The graphs below compare the population distribution with the sampling distribution of the mean for different sample sizes.

Small sample size ($n = 5$)



Large sample size ($n = 30$)



- Describe the shape of the sampling distribution when the sample size is small.

- Describe the shape of the sampling distribution when the sample size is large.

- What can you conclude about the shape of the sampling distribution as the sample size increases?



Key Idea

As the sample size increases, the sampling distribution of the mean becomes more symmetric and bell-shaped, regardless of the shape of the population.



A. HIGHER ORDER THINKING (HOTS)

Answer the following questions. Show your reasoning.

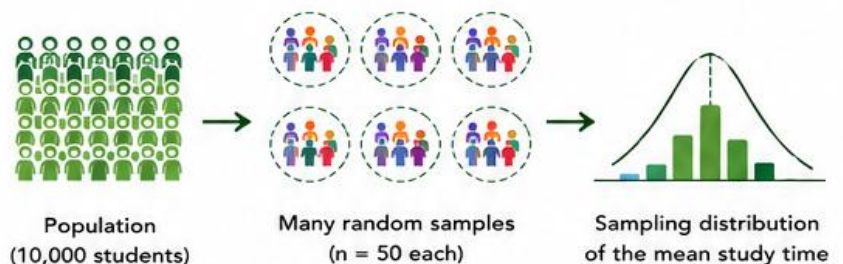
- 1 A population of exam scores is highly right-skewed. Explain why the sampling distribution of the mean becomes approximately normal as the sample size increases.
- 2 The population mean is $\mu = 75$. A researcher takes many samples of size $n = 25$ and obtains the sampling distribution of the mean. Why is the standard error (standard deviation of sample means) smaller than the population standard deviation?
- 3 If the sample size is increased from $n = 10$ to $n = 100$, what happens to the variability (spread) of the sampling distribution of the mean? Explain.
- 4 Why is understanding the sampling distribution important when making decisions or inferences about a population based on sample data?



B. APPLICATION (REAL-LIFE CONNECTION)

Read the scenario below and answer the questions.

A university wants to estimate the average number of hours students spend studying per week. The registrar knows that surveying all 10,000 students is not possible. Instead, the university decides to take many random samples of 50 students and compute the average study time for each sample.



- 1 What is the population in this study? _____
- 2 What is the statistic being collected from each sample? _____
- 3 How will the sampling distribution help the university make better decisions? _____



C. REFLECTION

Take a few minutes to reflect on what you have learned in this topic.

- 1 What did you learn from this topic that you did not know before?



- 2 Which part of this topic was most challenging for you? Why?

